Pattern Recognition, Vol. 25, No. 12, pp. 1459–1477, 1992
Printed in Great Britain

0031-3203/92 $5.00 + .00
Pergamon Press Ltd

ADDRESS BLOCK LOCATION ON ENVELOPES USING GABOR FILTERS

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(*Received* 25 *November* 1991; *in revised form* 17 *March* 1992; *received for publication* 18 *May* 1992)

Abstract—The large volume of mail and the increased cost of handling it has made postal automation an important domain for pattern recognition and computer vision research. A substantial amount of work is being done to design an automatic mail sorting system which can read and interpret the destination address on a mail piece and direct it to the appropriate bin. Robust optical character recognition (OCR) systems are now available which can read printed characters with great accuracy (> 99%). But, in order to read the destination address, the region in the image containing the address must first be located. Even though several approaches to address block location have been proposed in the literature, it remains a difficult problem. A simple method is presented for automatically identifying regions in envelope images which are candidates for being the destination address. The envelope image is considered to contain different textured regions, one of which corresponds to the text-content in the image. Thus, a texture-based segmentation method is used to identify the regions of text in the image. The method for texture discrimination is based on Gabor filters which have been successfully used earlier for a variety of texture classification and segmentation tasks. It is shown that only a small number of even-symmetric Gabor filters are needed in this application. The success of the texture-based segmentation algorithm for identifying address blocks is demonstrated on a number of test images. These results also demonstrate the invariance of the method to the orientation of text in the envelope image and the variations in the size and font of the text.

Postal automation   Address block location   Envelope images   Gabor filters   Texture
Segmentation

1. INTRODUCTION

The tremendous growth in the volume of paper mail and the increase in labor costs make the problem of postal automation increasingly important. One solution to this problem is a machine vision system which scans each mail piece in turn, reads the address and directs the mail piece to the appropriate bin. The availability of sophisticated optical character recognition (OCR) systems has effectively solved the problem of reading the machine-printed and, to a large extent, even the handprinted addresses. The problem of locating the destination address on the mail piece is still a major bottleneck in the development of an automatic mail sorting system. Consequently, this has become an important topic of current research in postal automation.

The physical attributes of mail pieces vary greatly. Even paper mail—letters and envelopes—comes in different sizes varying from the standard size for personal letters to large envelopes containing commercial mail. Often the envelope face contains text and graphic messages in addition to the stamps, postmarks, and destination and return addresses. To complicate the problem further, there is no fixed position for the destination address on the envelopes. Newspapers and magazines delivered by mail present these problems in the severest form.

We present an automatic method for locating address blocks on letters and envelopes. Our domain is currently limited to letters and envelopes which do not contain text messages other than the address blocks. We consider the envelope image to consist of several different types of textured regions, one of which results from the text-content in the image. Thus, we pose the problem of locating address blocks in envelope images as a texture discrimination problem. Our method for texture discrimination is based on Gabor filters which have been used earlier for a variety of texture classification and segmentation tasks. We have used a subset of the Gabor filters proposed by Jain and Farrokhnia.[1] Our segmentation algorithm for address block location is unsupervised and does not require user-specified thresholds and parameters. An additional advantage of our approach is that it is invariant to rotation of the mail pieces and appears to be robust to the font type and size of the printed handwritten text.

This paper is organized as follows. Section 2 briefly describes several approaches to address block location reported in the literature. Section 3 introduces the well-known multichannel filtering method and, in particular, Gabor filters for texture-based segmentation. In Section 4, we justify that the problem of address block location can be viewed as a "texture segmentation" problem. Section 5 gives experimental results which demonstrate the feasibility of the proposed method. Finally, Section 6 presents a summary of our work and a few concluding remarks.

2. APPROACHES TO ADDRESS BLOCK LOCATION

Several researchers have dealt with the problem of locating address blocks in images of mail pieces.[2-5] Expert systems utilizing AI techniques have been built for locating address blocks in the presence of other text and graphic information on the face of the mail piece. Another popular approach is to extract simple geometric features which characterize address blocks. This section contains a brief summary of these approaches reported in the literature.

2.1. *An expert systems approach*

An expert system for automatic sorting of mail pieces has been developed by Wang and co-workers.[3,4] In this scheme, the address blocks are located by the address block location subsystem (ABLS). The system is capable of using multispectral images (gray scale, color, infrared, or color under ultraviolet illumination). The ABLS consists of six major components:

(i) *Mail statistical database* (MSD): contains statistics of the geometric features of address labels on various mail items. This database has been generated after careful analysis of a large number of mail pieces.

(ii) *Tool box*: contains image processing tools for thresholding, segmentation, labeling connected components, discrimination between printed and handwritten text, merging and splitting of blocks, and analysis of the layouts and features of the text blocks.

(iii) *Rule-based inference engine*: acts as an interpreter for rules associated with each tool and performs forward or backward reasoning on different rule modules.

(iv) *Control system*: selects the appropriate tools from the toolbox. The utility of each tool in the current context is estimated and the tool with the highest utility is selected. The control system also combines new evidence, updates the context, and checks termination conditions.

(v) *Control data*: used by the control system to get information about interdependencies between tools and criteria for accepting a block as the destination address block.

(vi) *Blackboard*: contains the geometric attributes of blocks obtained after processing different types of images, the current context, and the confidence levels for the labeling hypothesis.

The ABLS may use one or more of the several image processing tools to process the input images of the mail piece. The evidence gathered from these tools is accumulated or integrated using a blackboard system. Each piece of evidence is accompanied by a confidence value which is a measure of the degree to which the evidence supports the hypothesis that a particular block is the destination address block (DAB). Based on the evidence available, a score is assigned to each candidate DAB. Finally, the candidate with the highest score is selected as the DAB.

A database of over 3000 mail pieces was used to develop the rules associated with the tools. Experiments on 174 mail pieces showed that the method was successful in locating the correct DAB in 81% of the cases. In 4% of the cases, the DAB was correctly identified but the DAB itself did not have enough information to automatically sort the mail piece. The method identified the wrong candidate block as the DAB in 10% of the cases and in the remaining 5% of the experiments, the method was unable to decide on any one candidate as the DAB. The average processing time in the above 174 experiments was 10.4 min on a Sun-3 system.[4]

2.2. *Address block location using geometric features*

Another technique for locating address blocks, proposed by Downton and Leedham,[2] uses geometric attributes to identify the most likely candidate for the destination address block. First, the input gray level image is thresholded to obtain a binary image. Next, horizontal "smearing" is used to identify regions of high "black" pixel density. The smearing value is chosen based on typical inter-word spacing, such that the words in each line of the address get connected. The smeared image is subsampled to reduce the image size from 2048×512 pixels to 256×64 pixels. For every adjacent block of 8×8 pixels in the smeared image, the corresponding pixel in the output reduced image is black if the input block contains more than 30 black pixels. (This threshold was determined empirically.) This reduction in resolution removes small regions of high black pixel density which are unlikely to be part of the address block. The eight-connected black blocks are grouped together in the low resolution image to form separate regions. The following features are computed for each of these regions:

(i) coordinates of the bounding rectangles for each region,

(ii) aspect ratio of each region,

(iii) density of black pixels in the smeared full-resolution image, and

(iv) density of black pixels in the input image.

These features are used to identify regions which are candidates for being part of the destination address.

Removal of regions which do not conform to the required thresholds for the above features reduces the time required for later processing. The remaining regions are smeared vertically to merge lines in the candidate regions. This time, the smearing value is selected based on the typical interline spacing. Again, all regions less than 100 pixels in height or 48 pixels in width or having aspect ratio (the ratio of width of a region to its height) greater than 35 are rejected. Finally, the block which is closest to a predetermined nominal point in the input image is selected. The nominal point is selected such that it is halfway down along the length axis and one-third down along the height axis.

Downton and Leedham[2] tried their method on a number of different types of envelope images. For envelopes with machine-printed characters, the method correctly identified the destination address block in about 98% of the cases. The performance of the method was not satisfactory for envelopes with hand-printed text. In many such cases, the method either failed to identify the destination address block correctly or only a part of the destination address block was identified. An obvious limitation of this approach is that envelope images with arbitrary orientations of text will not be processed correctly.

Yeh *et al.*[5] have proposed a method for locating address blocks which also utilize geometric features. This method applies the Laplacian operator, followed by a connected component analysis to identify each individual eight-connected set of pixels. This produces a very large number of connected components, many of them spurious in nature. A merging process is used to eliminate the undesired connected components. This process merges adjacent components if the relative contrast between them is low. For each of the remaining components, the following features are computed:

(i) position,
(ii) dimensions,
(iii) average gray level,
(iv) thickness, and
(v) "darkness vote" of the component.

The darkness vote determines the likelihood that the component is "darker" than its background.

The connected components are grouped based on the similarities of their geometric features. A *proximity tree* is constructed, which represents the similarity between connected components. The proximity tree is essentially a single-link dendrogram used in hierarchical clustering.[6] Cutting the tree will produce several different clusters of components. Each cluster is considered to belong to one of the following categories: destination address, return address, and stamp. Weighting functions corresponding to the different categories are used to assign scores to each cluster, based on its position. For example, the closer the cluster is to the top right corner, the higher its score for the category "stamp". Finally, the cluster with the highest score for belonging to the "destination address" category is selected as the destination address block.

3. MULTI-CHANNEL FILTERING FOR TEXTURE ANALYSIS

Most image textures can be characterized based on the local spatial frequency and orientation information present in the image. This is certainly true of the texture formed by text lines in envelope images. A variety of texture analysis techniques have been proposed in the image processing and computer vision literature.[7] Multichannel filtering techniques decompose an input image into a number of filtered images, each of which contains intensity variations over a narrow range of frequency and orientation. The Gabor filters have received considerable attention in the vision literature because the characteristics of certain cells in the visual cortex of some mammals can be approximated by these filters. In addition, these filters have been shown to possess optimal localization properties in both spatial- and frequency-domain and thus are well suited for texture segmentation problems.[1,8-11]

Gabor transforms allow one to obtain local estimates of frequency content.[12] A two-dimensional Gabor filter can be viewed as a sinusoidal plane of particular frequency and orientation, modulated by a Gaussian envelope. In spatial domain, the two-dimensional Gabor filter is given by

$$h(x,y)=\exp\left\{-\frac{1}{2}\left[\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right]\right\}\cos(2\pi u_0 x+\phi)$$

where σ_x and σ_y are the standard deviations of the Gaussian envelope along the x- and y-directions, respectively, and u_0 and ϕ are the frequency and phase of the sinusoidal plane wave along the x-direction (0° orientation). A rotation of the x–y plane by an angle θ will result in a Gabor filter at orientation θ.

The frequency- and orientation-selective characteristics are more obvious in the modulation transfer function (MTF) of the Gabor filter, $H(u,v)$, given by the Fourier transform of $h(x,y)$. For $\phi = 0°$ (even-symmetric filters):

$$H(u,v)=A\left(\exp\left\{-\frac{1}{2}\left[\frac{(u-u_0)^2}{\sigma_u^2}+\frac{v^2}{\sigma_v^2}\right]\right\}+\exp\left\{-\frac{1}{2}\left[\frac{(u+u_0)^2}{\sigma_u^2}+\frac{v^2}{\sigma_v^2}\right]\right\}\right)$$

where $\sigma_u = 1/2\pi\sigma_x$, $\sigma_v = 1/2\pi\sigma_y$, and $A = 2\pi\sigma_x\sigma_y$.

Figure 1 shows the MTFs of eight Gabor filters which have been used in this study. These filters have the characteristics of band-pass filters.[13]

An unsupervised texture segmentation algorithm using Gabor filters has been presented by Farrokhnia,[13] and by Jain and Farrokhnia.[1] The main steps of the algorithm are given below.

(i) Filter the input image through a bank of n even-symmetric Gabor filters, to obtain n filtered images.

(ii) Compute the feature images consisting of the "local energy" estimates over Gaussian windows of appropriate size centered at every pixel in each of the filtered images. The actual method for computing the feature images is described later.

(iii) Cluster the feature vectors corresponding to each pixel using a squared-error clustering algorithm to obtain a segmentation of the input image into K clusters. Use the row and column pixel coordinates as additional features for clustering to obtain homogeneous regions with smooth boundaries.

For an image with N columns, where N is a power of 2, the Gabor filters with radial frequencies

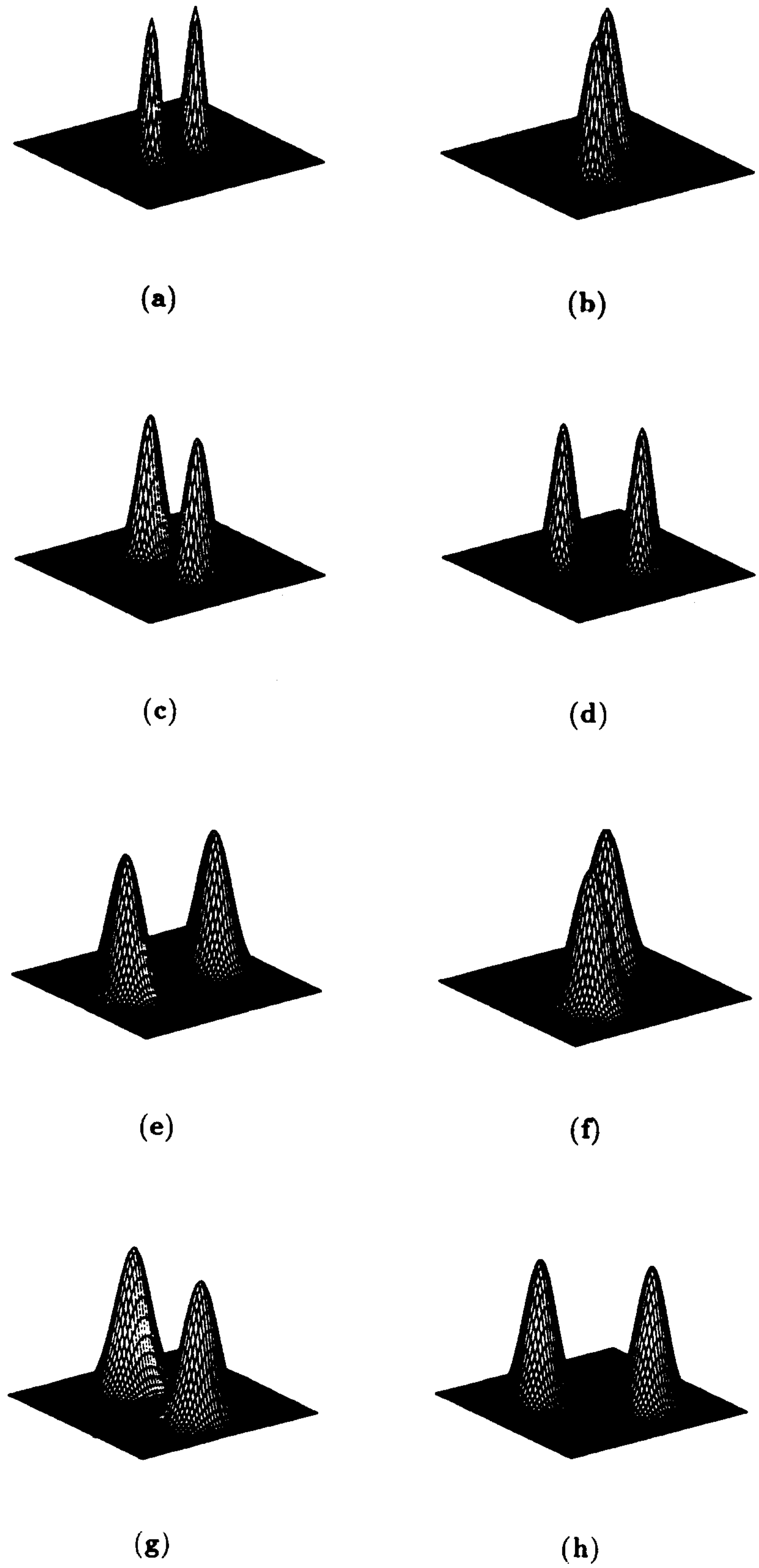


Fig. 1. Perspective plots of the MTFs of the eight Gabor filters used in our experiments. For a 256 × 256 image, the filter parameters are as follows: (a) $u_0 = 32\sqrt{2}$, $\theta = 0°$; (b) $u_0 = 32\sqrt{2}$, $\theta = 45°$; (c) $u_0 = 32\sqrt{2}$, $\theta = 90°$; (d) $u_0 = 32\sqrt{2}$, $\theta = 135°$; (e) $u_0 = 64\sqrt{2}$, $\theta = 0°$; (f) $u_0 = 64\sqrt{2}$, $\theta = 45°$; (g) $u_0 = 64\sqrt{2}$, $\theta = 90°$; (h) $u_0 = 64\sqrt{2}$, $\theta = 135°$.

$1\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, \ldots, (N/4)\sqrt{2}$ cycles/image are selected. For each radial frequency, u_0, four filters, with orientations, $\theta = 0°$, 45°, 90°, 135°, are used.[1] Thus, for an image with $N = 256$, a bank of 28 filters ($n = 4 \times 7$) would be used.

To specify a Gabor filter completely, we must not only specify its radial frequency and direction of the sinusoid, but also the spread of the Gaussian that modulates it. In the spatial domain the spread of the two-dimensional Gaussian is given by its standard deviations σ_x and σ_y. The corresponding parameters in the frequency domain are $\sigma_u = 1/2\pi\sigma_x$ and $\sigma_v = 1/2\pi\sigma_y$. However, in the frequency domain, it is more meaningful to describe the parameters σ_u and σ_v in terms of the half-peak magnitudes of the frequency bandwidth, B_r (in octaves), and the orientation bandwidth, B_θ (in deg).[13] Using B_r and B_θ, σ_u and σ_v can be computed as follows:

$$\sigma_u = \frac{u_0}{\sqrt{2}} \frac{2^{B_r} - 1}{2^{B_r} + 1}$$

and

$$\sigma_v = \frac{u_0}{\sqrt{2}} \tan\left(\frac{B_\theta}{2}\right).$$

In our experiments, $B_r = 1$ octave and $B_\theta = 45°$.

Before generating the feature images, a nonlinearity is introduced into each filtered image by applying the following transformation:[1,13]

$$\psi(t) = \tanh(\alpha t) = \frac{1 - e^{-2\alpha t}}{1 + e^{-2\alpha t}}.$$

For $\alpha = 0.25$, this function acts almost as a thresholding transformation, and serves as a blob-detector in the filtered images. The output of this non-linear stage is a set of *response images*. We define a measure of the texture energy, called the average absolute deviation (AAD), which is the mean response value over a small window centered at each pixel. More formally, the feature image $e_k(x, y)$ corresponding to the filtered image $r_k(x, y)$ is given by

$$e_k(x, y) = \frac{1}{M^2} \sum_{(a,b) \in W_{xy}} |\psi(r_k(a, b))|$$

where W_{xy} is a window of size $M \times M$ centered at pixel (x, y) in the filtered image. As described in reference (1), W_{xy} is a Gaussian window with standard deviation $\sigma = 0.5\,T$ where $T = N/u_0$ pixels.

The n feature images are normalized to zero mean and unit standard deviation to prevent a feature with higher numerical range dominating the others. The normalized spatial coordinates of every pixel are also used in the clustering process. These features are used to smooth out the cluster boundaries and to delete very small noisy connected components. The values in the n feature images corresponding to a pixel together with its row and column coordinates define a $(n + 2)$-dimensional feature vector. The normalized features are clustered by the squared-error clustering algorithm, CLUSTER.[6] In practice, due to the large number of pixels (and hence patterns) in the image, clustering becomes impractical (for a 256×512 image, a total of 131,072 patterns would have to be clustered). We circumvent this problem by clustering only a subset of randomly chosen patterns (pixels) from the image. Specifically, for a 256×512 image, 8000 randomly chosen pixels (patterns) are clustered. The patterns in these clusters are then used for training a linear (minimum Euclidean distance) classifier to classify all the pixels in the image into K classes.

4. FILTERING BASED METHOD FOR LOCATING ADDRESS BLOCKS

The proposed method for locating address blocks is based on a multi-channel filtering approach to texture segmentation.[1,13] The text in the envelope image is considered to define a textured region. Non-text components in the envelope image, including blank spaces, are considered as regions with different texture types. Thus, the problem of locating the text in the envelope image can be posed as a texture segmentation problem. A three-cluster solution ($K = 3$) identifies the regions consisting of printed text, non-text (blank spaces and regions which have slow intensity variations), and the boundaries between the text regions and the non-text regions. We utilize a subset of the filter bank used earlier in texture segmentation studies[1,13] for identifying the destination address block in the envelope image.

Based on our initial experiments with several document images, we decided to use only the eight filters corresponding to the two highest radial frequencies (see Fig. 1). Using only the higher frequency filters lays more emphasis on identifying regions corresponding to high-frequency texture typical of the printed text. Figure 2 shows the image of an envelope face. The eight filtered images, are shown in Fig. 3. These filtered images clearly show that filter responses in the text regions are different from those in the non-text regions. The corresponding feature images are shown in Fig. 4. Note that the feature values corresponding to the text regions are consistently high. Figure 5 shows the plots of the mean feature values of the eight filter-features computed for the input image shown in Fig. 2, for each of the three clusters. The highest mean feature values correspond to cluster number 2, the text cluster.

It has been experimentally ascertained that the three-cluster segmentation gives the desired separation of text from non-text regions. We can now isolate the text regions in the envelope image by applying a mask to the original image which will pick out only those pixels that correspond to the cluster representing the text. After isolating the segment or cluster corresponding to the text, a connected component analysis is used to identify the individual "words" in the text. Since each word is expected to be associated with one connected component, components smaller than 30

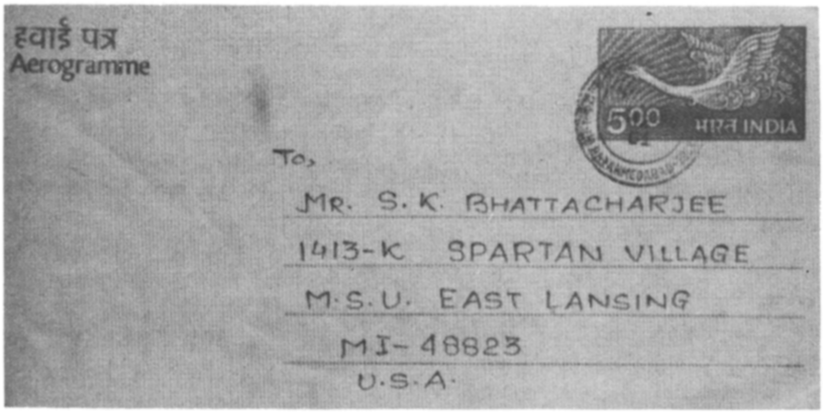
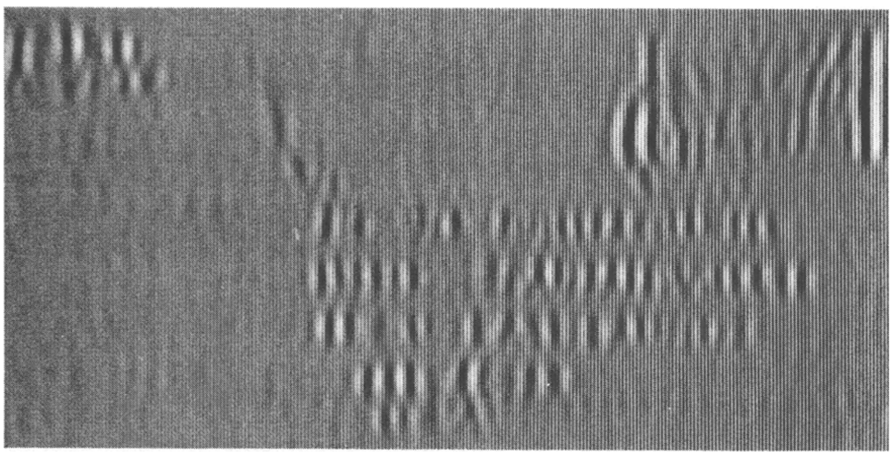
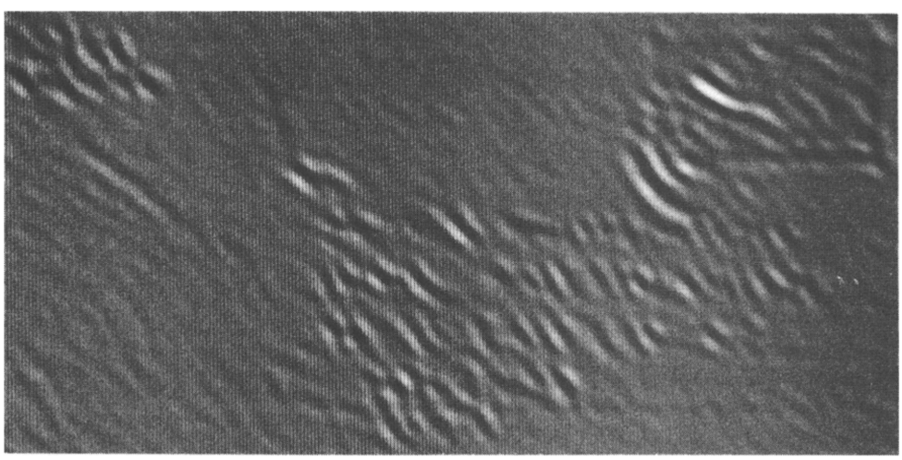


Fig. 2. 256 × 512 images of an air-mail letter scanned at a resolution of 75 dpi on a flat-bed scanner.

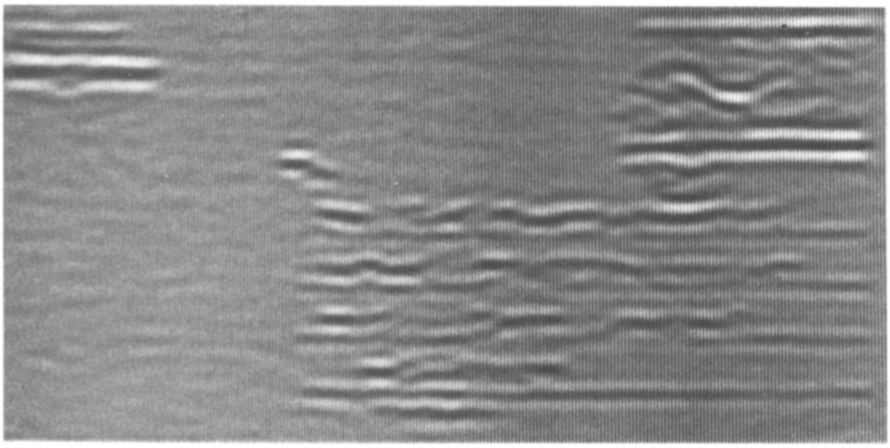


(a)

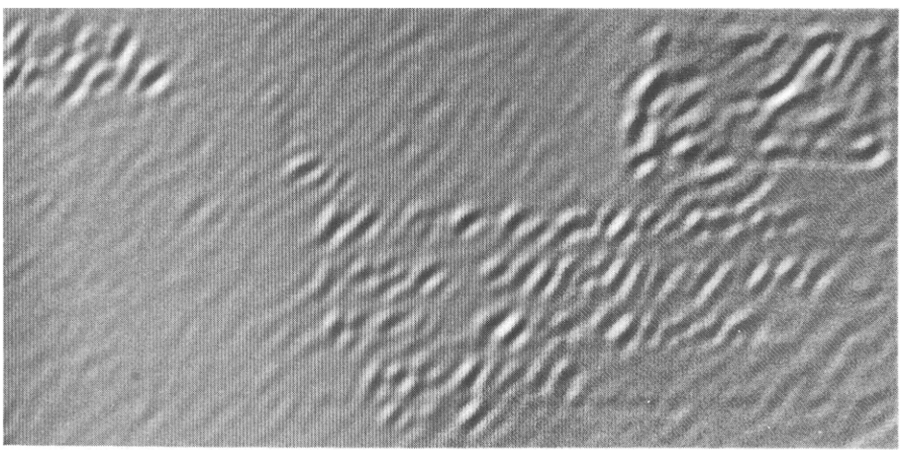


(b)

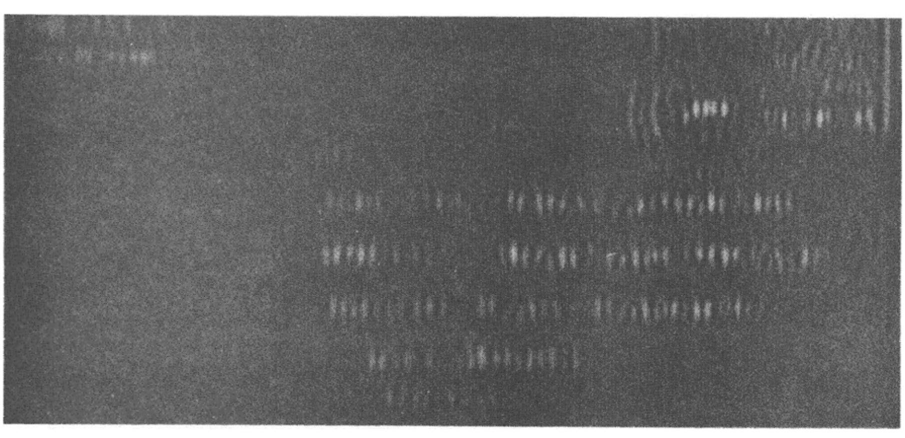
Fig. 3. Results of filtering the envelope image in Fig. 2. The frequency u_0 and orientation θ of the filters are: (a) $u_0 = 64\sqrt{2}, \theta = 0°$; (b) $u_0 = 64\sqrt{2}, \theta = 45°$; (c) $u_0 = 64\sqrt{2}, \theta = 90°$; (d) $u_0 = 64\sqrt{2}, \theta = 135°$; (e) $u_0 = 128\sqrt{2}, \theta = 0°$; (f) $u_0 = 128\sqrt{2}, \theta = 45°$; (g) $u_0 = 128\sqrt{2}, \theta = 90°$; (h) $u_0 = 128\sqrt{2}, \theta = 135°$.



(c)

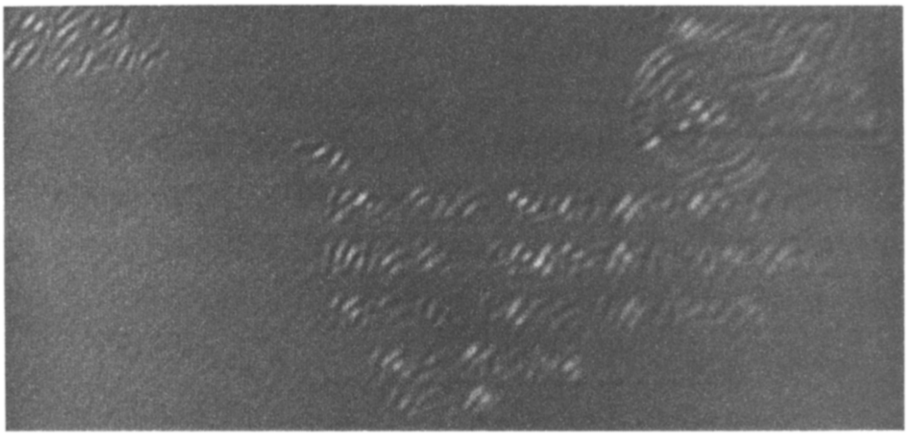


(d)

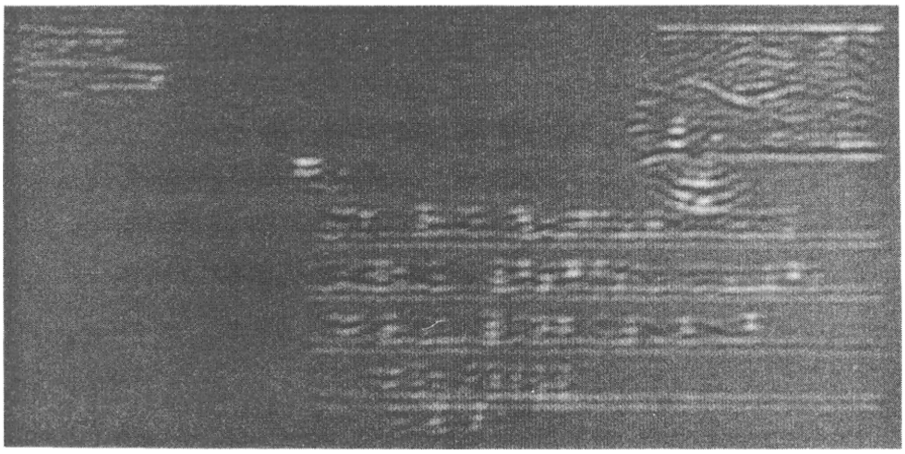


(e)

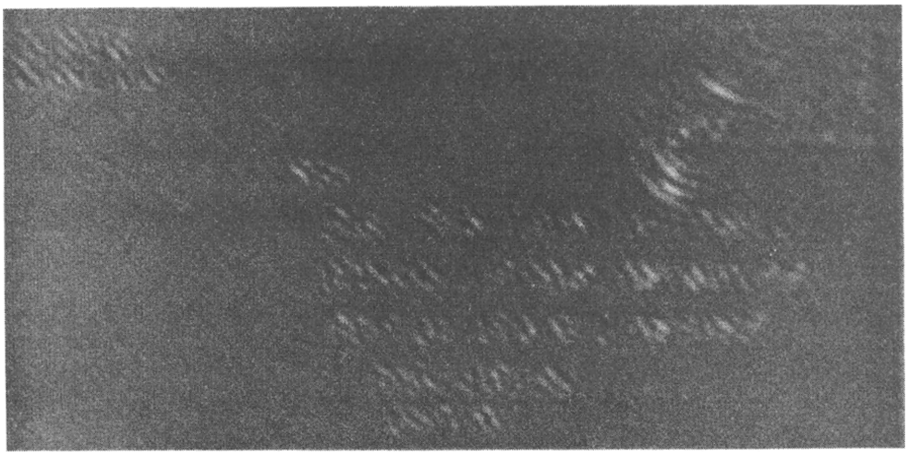
Fig. 3. (*Continued.*)



(f)

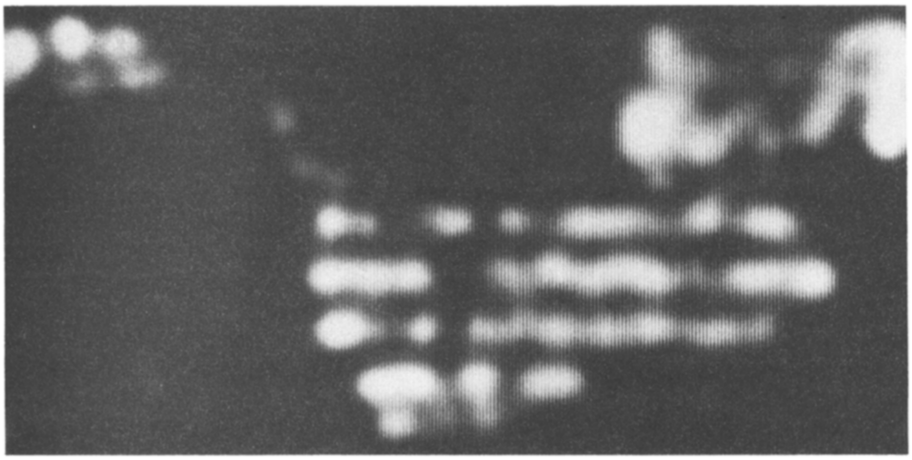


(g)

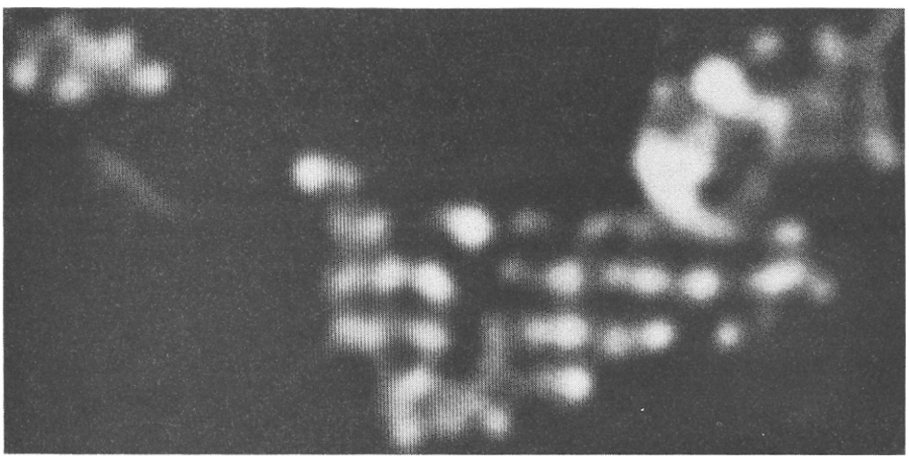


(h)

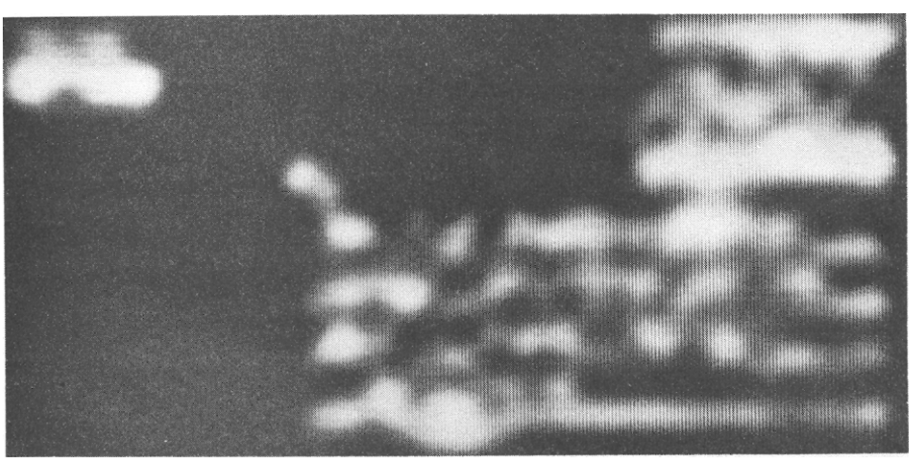
Fig. 3. (*Continued.*)



(a)

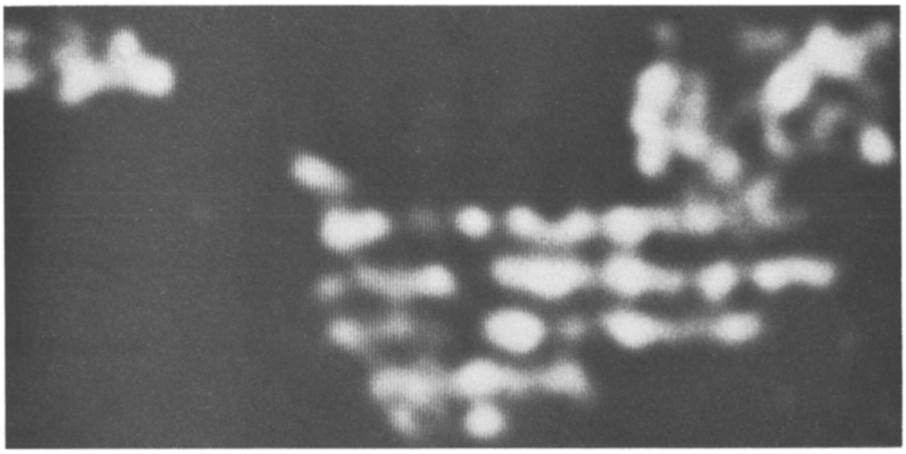


(b)

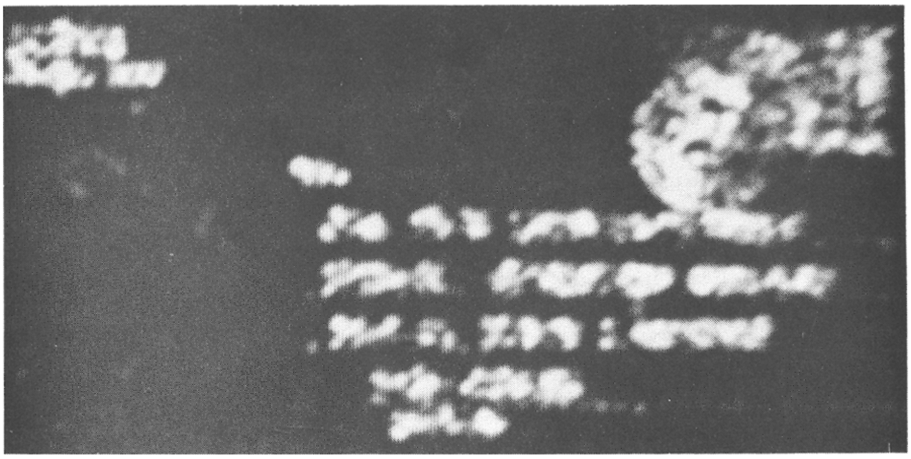


(c)

Fig. 4. Eight feature images for the envelope image shown in Fig. 2. Images (a)–(h) are obtained from filtered images shown in Figs 3(a)–(h), respectively.



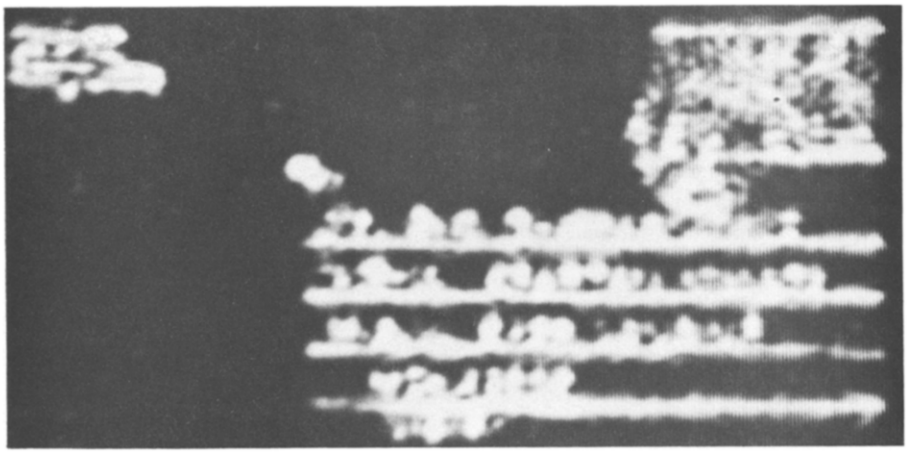
(d)



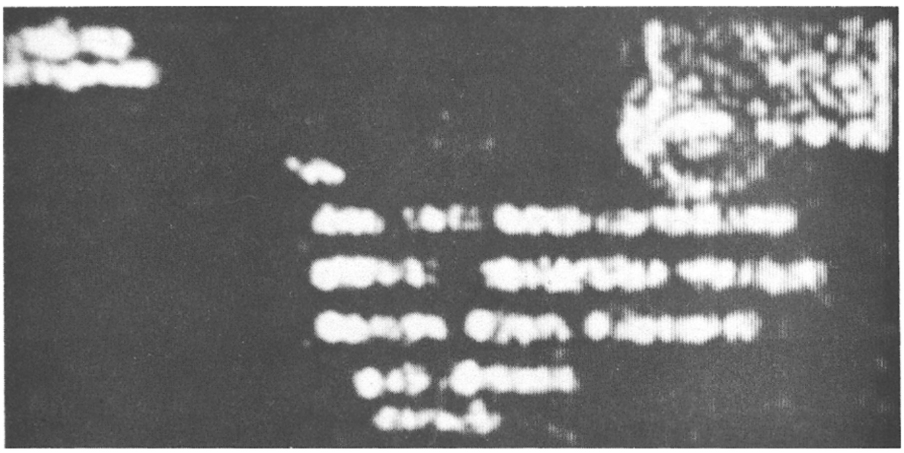
(e)

(f)

Fig. 4. (*Continued.*)



(g)



(h)

Fig. 4. (*Continued.*)

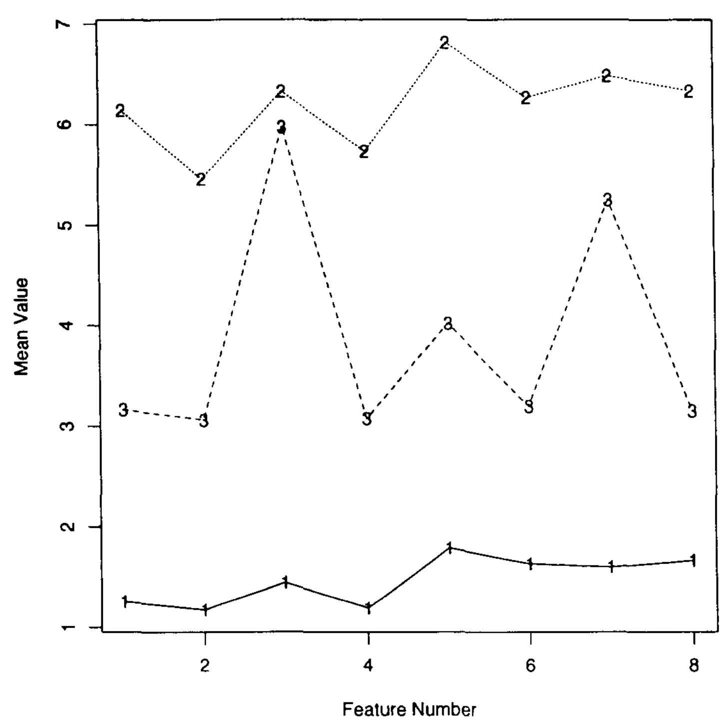


Fig. 5. Mean feature values of the eight Gabor filter features corresponding to the image in Fig. 2 for the 3-cluster segmentation. The label on each line in the plot is the segment number represented by the respective line (segment No. 1, non-text; segment No. 2, text; segment No. 3, boundary regions between the non-text and the text).

pixels in area or larger than 1500 pixels in area are removed. Such components are either too small or too large to be candidates for words in a postal address. The connected components which satisfy the size constraints represent potential words in the address blocks in the envelope image. Note that the segmentation algorithm described above is invariant to rotation of the envelope image and will properly identify the text even if the text is not aligned with the horizontal and vertical directions. The next step is to determine the orientation (with respect to the x-axis) of the text in the envelope image. This is necessary for grouping the words to form lines and to further group the lines to form blocks of text which may be potential address blocks. A two-dimensional principal component analysis of the (x, y) coordinates of the pixels in a connected component allows us to determine the major and minor directions of the component. In order to find a common coordinate system for all the connected components, a histogram of the major directions of all the connected components is constructed. The mode of this histogram is selected as the common major axis for all the connected components. The common minor axis can be computed as the direction orthogonal to the common major axis.

To identify the blocks of text in the envelope image, the following procedure is followed. First, the pixel coordinates in all the connected components are projected onto the minor axis. A one-dimensional histogram is constructed for this projection. This histogram typically consists of a number of the clusters of peaks. Each cluster corresponds to a group of text lines close to one another in the envelope image. Thus, by merging the histogram peaks that are less than a certain distance apart from one another, we can identify groups of lines. In our experiments, this distance threshold was chosen to be 15 units. Next, for each group of text lines so identified, the coordinates of the pixels in the connected components included in the group are projected on to the major axis. A one-dimensional histogram, similar to the one in the previous step, is constructed for the projected points. Peaks in this histogram which are close to one another are merged. This allows us to separate groups of connected components which may overlap on the minor axis. A high-level description of our algorithm for locating address blocks is given below:

(i) Obtain a 3-cluster segmentation of the input image.

(ii) Identify connected components belonging to the text regions.

(iii) Eliminate very small (< 30 pixels) and very large (> 1500 pixels) connected components.

(iv) For each connected component, perform a principal components analysis on the pixel coordinates to compute the major and minor axes of the connected component. Construct the smallest bounding rectangle for each connected component.

(v) Construct a histogram of the major axes. Select the direction corresponding to the histogram mode as the common major axis. The minor axis is orthogonal to the major axis.

(vi) Project the pixel coordinates of each connected component onto the minor axis to cluster lines of text. For each cluster of lines, project the pixel coordinates onto the major axis to identify groups of words.

(vii) For each group of words, construct the enclosing rectangle aligned with the coordinate system defined by the major and minor axes computed in step (v). Each rectangle is a candidate for being the destination address block.

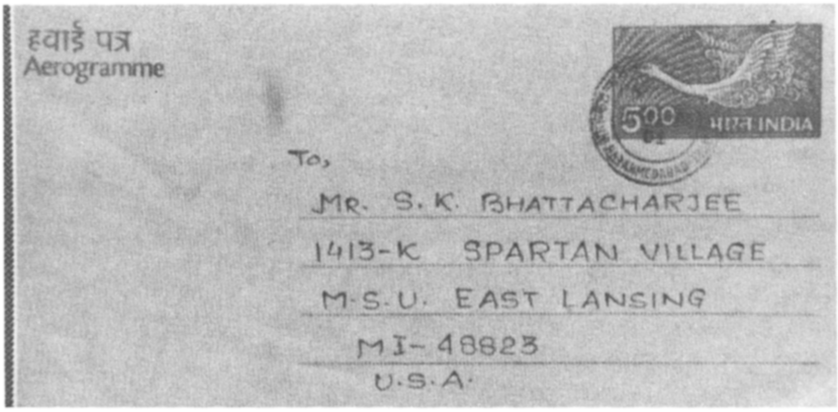
We have thus identified a set of text blocks in the envelope image. One of these blocks is the destination address. Heuristics ranging from simple ones involving the analysis of the positional relationships of the candidate blocks to more complex ones involving layout analysis of each identified block, can be used to label one of the blocks as the destination address block. We have not implemented any heuristic to identify the destination address block.

5. EXPERIMENTAL RESULTS

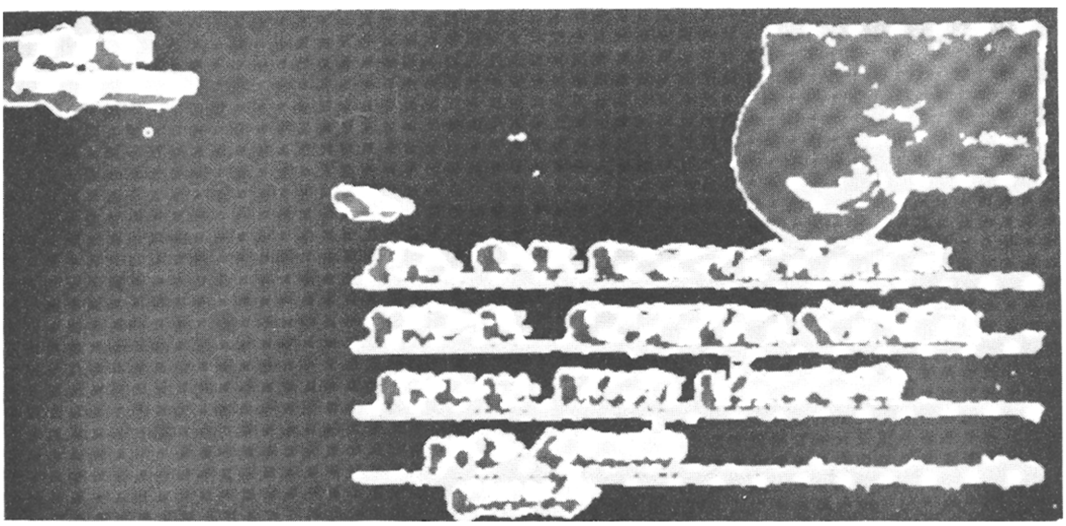
This section presents the results of the texture-based address block location scheme proposed in Section 4. The results are based on several images of air-mail letters and envelopes, all of conventional size. The images were digitized at a resolution of 75 dots per inch (dpi), using the Sharp JX-300 flat-bed scanner in our Pattern Recognition & Image Processing (PRIP) Laboratory. We report here results on three different envelope images, called MP-1, MP-2, and MP-3.

Figure 6(a) shows the 256×512 input image of MP-1, Fig. 6(b) shows the segmentation results for the three-cluster segmentation of the input image, and Fig. 6(c) shows the regions identified as text by the segmentation algorithm. The stamp on the air-mail letter is also included as text because the frequency content in that region is comparable to that in the text regions. The results of removing the spurious text components are shown in Fig. 6(d). Note that the stamp region and several small noisy segments are now eliminated. Figure 6(e) shows the enclosing rectangular boxes for each word and for the logical groups of words, superimposed on the input image. Two potential address blocks have been identified, one of which is the destination address block.

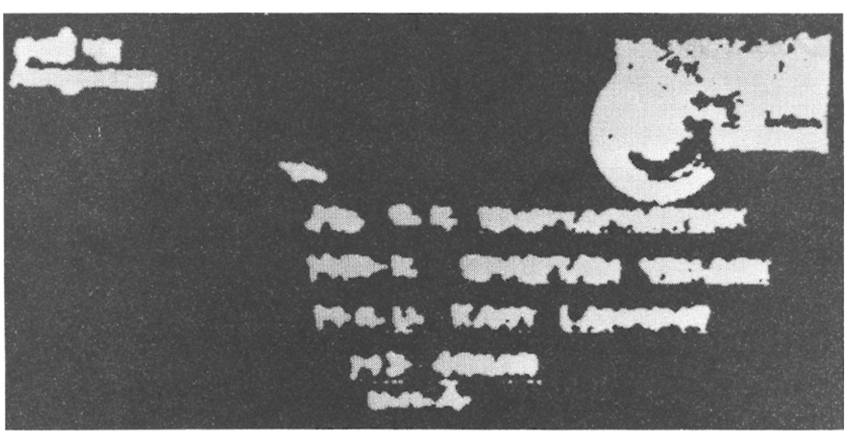
Figure 7 shows the results on the 256×512 image of an envelope (MP-2). The input image and the segmentation results are shown in Figs 7(a) and (b), respectively. As in the previous case, Fig. 7(c) shows the regions identified as text by the segmentation algorithm. Figure 7(d) contains the results after removing those connected components which do not satisfy the size constraints. Finally, the results of superimposing the bounding boxes around the individual words and around the clusters of words identified by the proposed method, are shown in Fig. 7(e). Again,



(a)

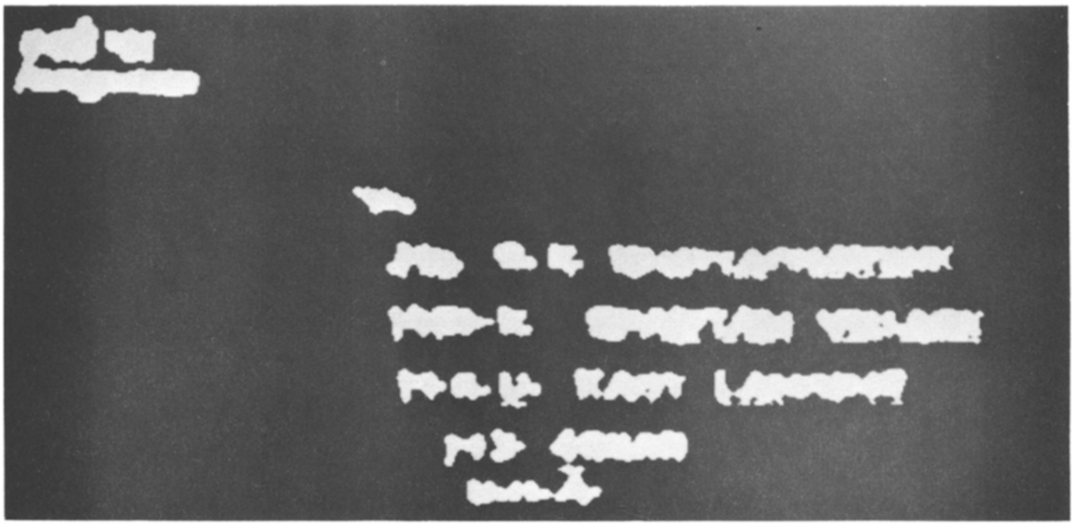


(b)

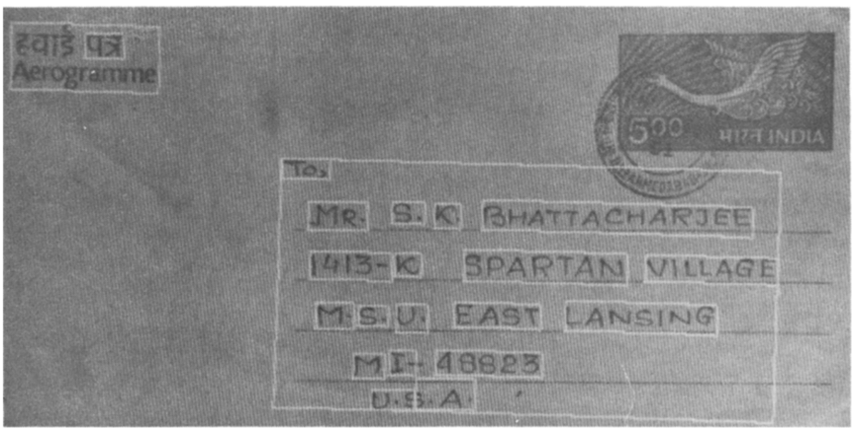


(c)

Fig. 6. Results of the address block location algorithm for the image of an air-mail letter (MP-1): (a) input image (256 × 512); (b) 3-cluster segmentation of (a); (c) the "text" regions as identified by the segmentation algorithm; (d) the result after eliminating the very large and very small connected components; (e) the enclosing boxes for the individual words as well as for the candidate address blocks, superimposed on the input image.

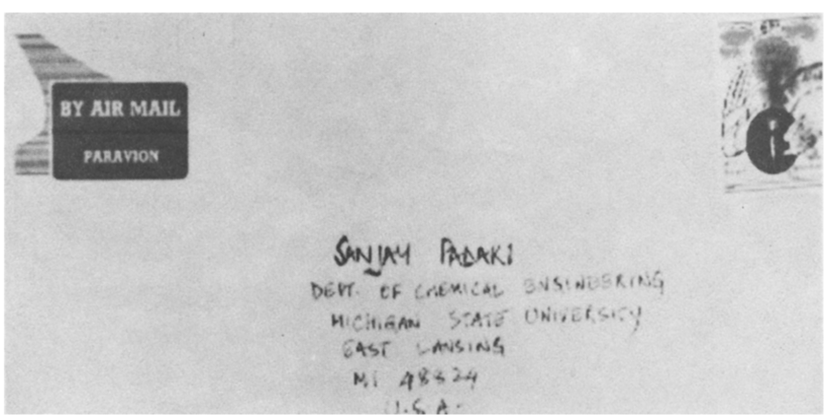


(d)



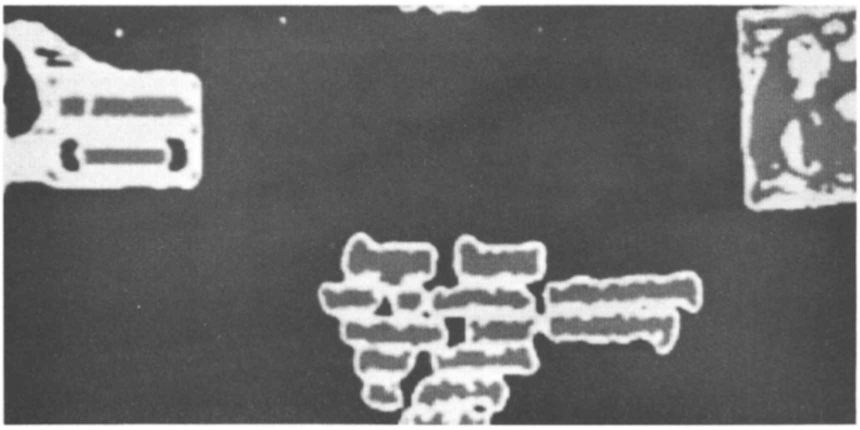
(e)

Fig. 6. (*Continued.*)

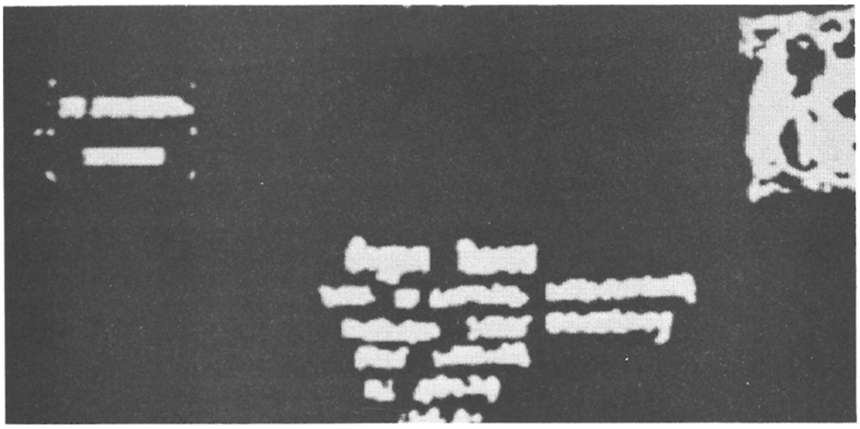


(a)

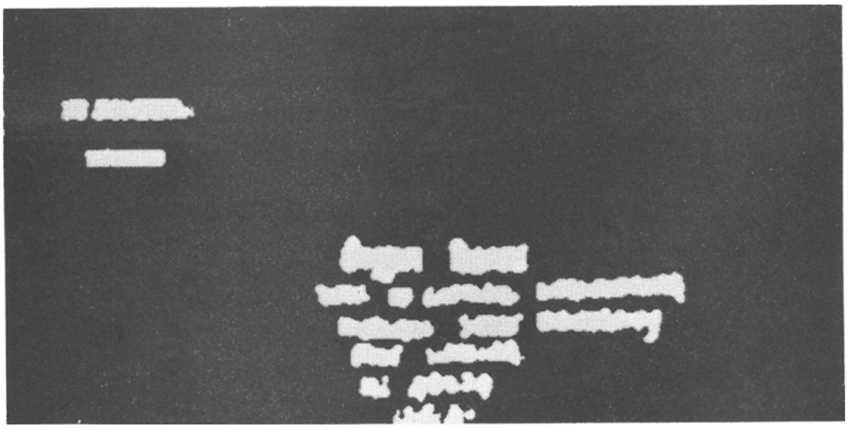
Fig. 7. Results of the address block location algorithm for the image of an air-mail envelope (MP-2): (a) input image (256 x 512); (b) 3-cluster segmentation of (a); (c) the "text" regions as identified by the segmentation algorithm; (d) the result after eliminating the very large and very small connected components; (e) the enclosing boxes for the individual words as well as for the candidate address blocks, superimposed on the input image.



(b)

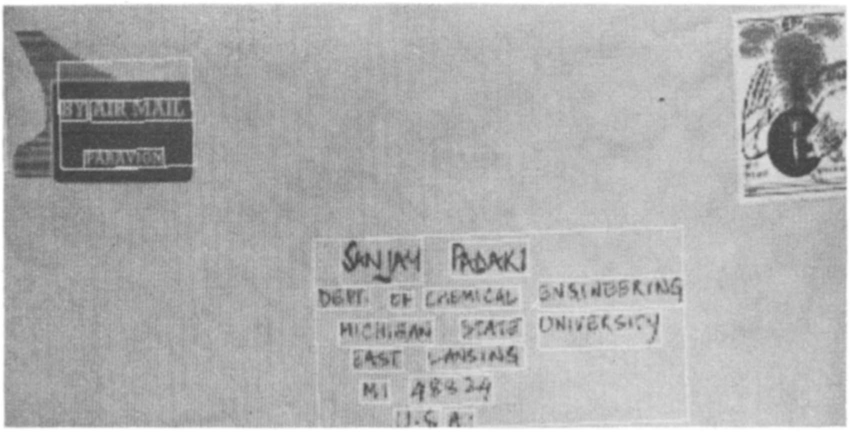


(c)



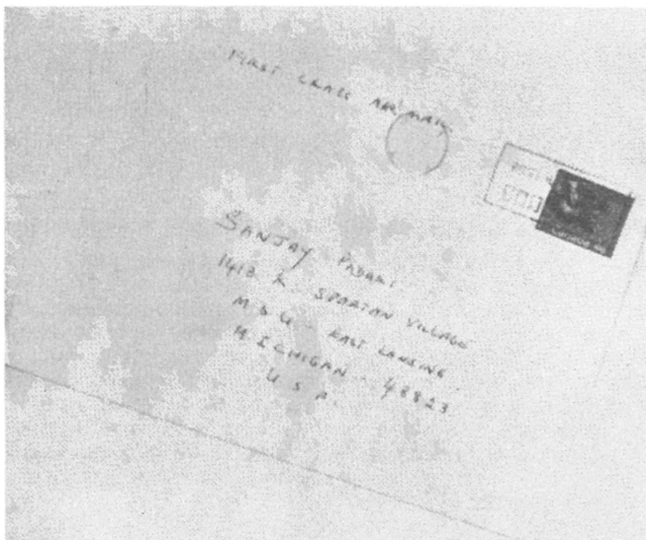
(d)

Fig. 7. (*Continued.*)

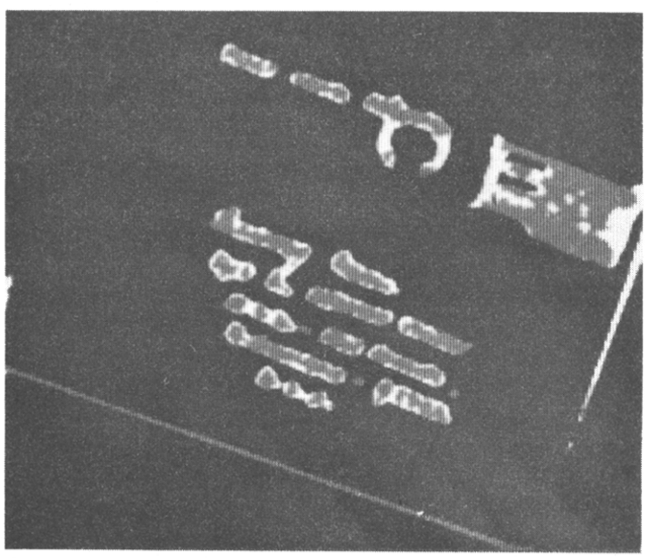


(e)

Fig. 7. (*Continued.*)

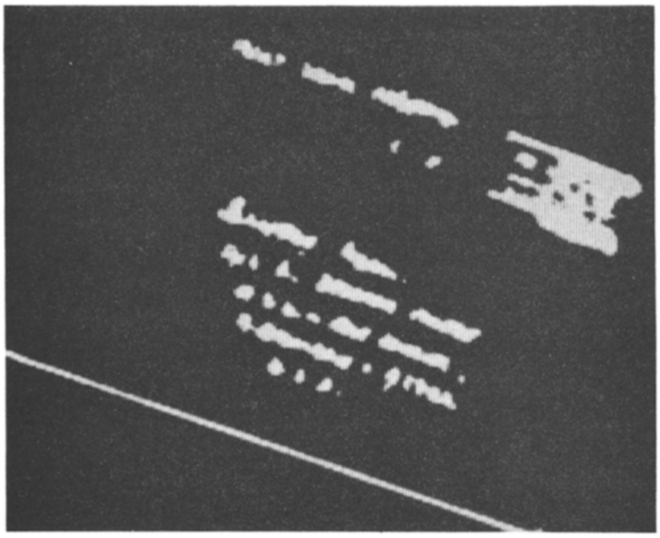


(a)

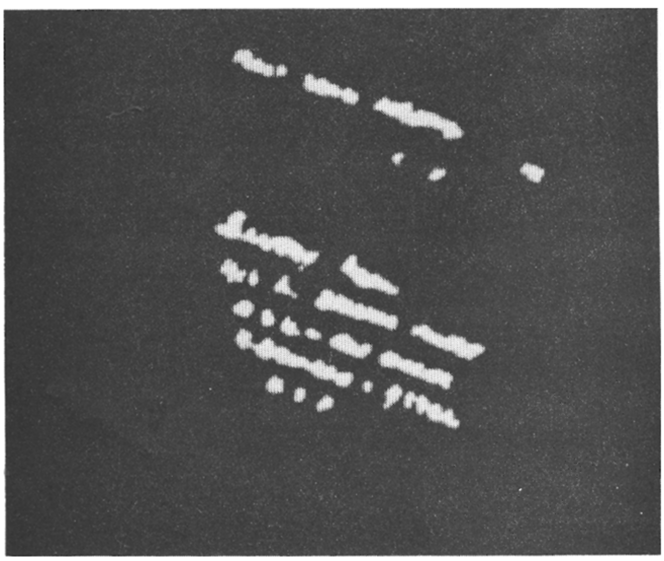


(b)

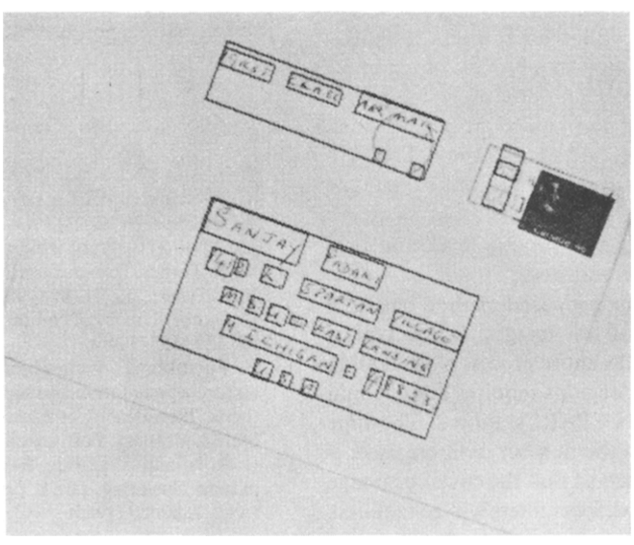
Fig. 8. Results of the address block location algorithm for the image of an envelope (MP-3) placed at an arbitrary angle while scanning: (a) input image (350 × 424); (b) 3-cluster segmentation of (a); (c) the "text" regions as identified by the segmentation algorithm; (d) the result after eliminating the very large and very small connected components; (e) the enclosing boxes for the individual words as well as for the candidate address blocks, superimposed on the input image.



(c)



(d)



(e)

Fig. 8. (*Continued.*)

two potential address blocks have been identified, one of which contains the destination address.

Finally, Fig. 8 demonstrates the ability of the proposed algorithm to process envelope images even if the envelopes are not appropriately aligned while scanning. The input image (MP-3) is shown in Fig. 8(a). Figure 8(b) shows the segmentation result and Fig. 8(c) shows the regions identified as text. The results after removing the spurious components are shown in Fig. 8(d). The bounding rectangles of the individual words and for the groups of words identified by the proposed algorithm are shown in Fig. 8(e), superimposed on the input image. In this example, three candidate blocks have been identified, one of which is the destination address block.

6. SUMMARY AND CONCLUSIONS

The task of postal automation continues to provide challenging research problems. The development of robust OCR systems has made automatic interpretation of addresses on mail pieces very viable. The stumbling block in the automatic sorting of mail is the correct location of the destination address. Several methods have been proposed to solve this problem. We have presented a new method for automatically locating address blocks on envelopes. The text on the face of the envelope is considered to define a texture, and a texture-based segmentation approach is used to identify the text regions in the envelope image. The proposed method uses general texture features which have been utilized earlier in a variety of texture classification and segmentation experiments. A texture segmentation algorithm based on Gabor filters is used to identify the text regions.

The method for address block location proposed in this paper has several advantages. Unlike other methods for address block location, we do not rely on gray-level thresholding to identify connected components. Our method is invariant to rotation that may be introduced in the image in the process of scanning. We use relatively low-resolution (75 dpi) images. At this resolution, our method can identify individual words as separate connected components. The proposed method is also robust to variations in the type and size of the fonts appearing in the input image. In addition, our algorithm can process both machine printed and handwritten addresses.

The performance of the proposed method has been demonstrated on several test images. For a typical image of size 256×512, the entire process of generating candidate address block regions requires about 1 min of CPU time on a SUN SPARCstation 2. The time requirement depends on the number of filters used. If the problem is constrained so that the envelope image is always correctly aligned, fewer filters will be required. Our current implementation performs the filtering in the frequency domain. This entails the computation of the Fourier transform and its inverse. The high-frequency filters that we are using can be represented as small sized convolution masks (9×9 and 11×11) in the spatial domain. Filtering in the spatial domain can be done efficiently using special hardware. The unsupervised clustering stage can also be implemented in hardware.[14] However, computational savings will be achieved if the clustering stage is replaced by supervised classification. The results of initial experiments using a two-class supervised classification (text vs. non-text) are encouraging. These modifications will reduce the computation time considerably.

Acknowledgement—This work was supported by the National Science Foundation under NSF grant CDA-88-06599.

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